Distributed Secondary Control for Current Sharing and Voltage Restoration in DC Microgrid

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***Abstract*—Direct current (DC) microgrid is being increasingly investigated in modern power grid. An important issue in DC microgrid operation is to ensure proper current sharing among converters. While this has been addressed through droop control, the resulting voltage deviation in DC bus has to be compensated. To solve this problem, a new distributed secondary control scheme is presented in this paper for both current sharing and voltage restoration. A key part of the presented scheme is the integration of a new parameter 'virtual voltage drop' defined from droop gain and line resistance. Since the DC bus voltage is not required as a feedback signal, the proposed secondary control is simple and easy to design and implement. In addition, as the proposed scheme has no requirement for the loads, it can handle both resistance loads and constant power loads (CPLs). Simulations as well as experimental studies are carried out to demonstrate the effectiveness of the proposed scheme.**

***Index Terms*—DC microgrid, current sharing, voltage restoration, distributed secondary control.**

I. Introduction

IN RECENT years, the concept of direct current (DC) microgrid is being increasingly investigated in modern power grid due to its simplicity, reliability and efficiency [1]–[3]. A large number of renewable energy sources, such as photovoltaic panels, have DC characteristics in nature. Thus, it is more convenient and efficient to connect the DC sources and DC loads directly to form a DC microgrid rather than an alternating current (AC) microgrid. Compared with AC microgrid in [4]–[7], DC microgrid has no reactive power and harmonics. Therefore, it provides higher power quality and is also easier to control [8]–[10].

In DC microgrid, it is an important control requirement to ensure proper current sharing for power dispatch. To meet this requirement, droop control has been widely adopted [11]–[14]. By setting the droop gain as a 'virtual resistance', which is sufficiently larger than the line resistance, droop control can ensure the current outputs from the DC sources to be proportional to each other as designed. Moreover, droop control is implemented in a fully decentralized way, and thus no communication is required among the DC sources.

However, as pointed out in [15], when a high droop gain is set in droop control, the DC bus voltage will deviate largely from its nominal value. This indicates that there exists a trade-off between the current sharing accuracy and DC bus voltage violation if only droop control is used. Therefore, complementary methods have to be designed to solve the voltage violation problem caused by droop control.

The idea of secondary control is proposed to complement droop control for both current sharing and voltage restoration in [16]. By integrating secondary control into droop control, proper current sharing and bus voltage regulation can be achieved simultaneously. In [16], a framework is presented for centralized proportional and integral secondary control. Though centralized control works well in ensuring satisfactory control performance, it has inherent limitations such as being sensitive to single-point failures [17]. References [18]–[19] have proposed decentralized output constrained control algorithms for single-bus DC microgrids. They can guarantee not only DC voltage convergence but also a bounded transient tracking error. However, these algorithms require each DC converter to have access to the DC bus voltage, and thus may fail to work when this requirement is not met.

To address the above challenges, distributed secondary control has been presented in the existing literature. In [20], the average value of the current from all DC sources is used to design the secondary control signal. Similarly, the work in [15] uses the average value of voltage for mitigating voltage violation. However, how to compute the average value is not

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discussed, and the controller is designed in a feed-forward manner. In [21]–[22], a framework involving designing a voltage and a current regulator is used to ensure voltage correction and load sharing. However, it fails to guarantee exact DC bus voltage restoration. In [23], a hybrid distributed secondary controller is proposed which consists of both a continuous-time part and a discrete-time part. This strategy ensures both current sharing and DC voltage regulation with a reduced communication burden. In [24] a distributed control strategy is proposed with the DC bus voltage as the feedback signal. Moreover, by adopting the idea of pinning control, the controller in [24] can be applied to the case where only one DC converter has access to the DC bus voltage signal. Then, the results are extended to the scenarios with communication limitations in [25]. However, the control strategies in [23]–[25] all require the DC bus voltage as a feedback signal, and it is not easy to set the control parameters because some complicated stability conditions are involved. In addition, they can only deal with resistance loads and the constant power loads (CPLs) are not considered.

In this paper, a new distributed secondary control scheme is presented for DC microgrid. It is able to not only guarantee proper current sharing, but also maintain the DC bus voltage at the nominal value. The contributions of this paper mainly lie in the following aspects:

- A new parameter 'virtual voltage drop' is defined from the droop gain and line resistance. It is shown that the average value of this new parameter plays a key role in proper current sharing and voltage restoration;
- For each converter, by only exchanging information with its neighbours, a new algorithm is designed to enable each converter to estimate the average value of the newly defined parameter. This forms our proposed distributed secondary control scheme;
- It is proved theoretically that the proposed distributed secondary control scheme can ensure both current sharing and voltage restoration. Compared with existing results, the proposed scheme has two advantageous features: 1) It is simpler and easier to implement as it does not require the DC bus voltage as the feedback signal. 2) It is able to handle both resistance loads and CPLs.

Both simulations and experiments are conducted to illustrate the effectiveness of the proposed control scheme.

The rest of this paper is organized as follows. The notations and symbols used in this paper are listed in Table I. Section II formulates the problem. Then, Section III presents our new distributed secondary control scheme. After that, five case studies are provided in Section IV. Experimental results are also presented in Section V. Finally, Section VI concludes the paper.

TABLE I
NOTATIONS AND SYMBOLS

Acronyms	
DC	Direct current
CPL	Constant power load
PWM	Pulse width modulation
Symbols of the DC microgrid	
i	Index for DC converters, $i = 1, \cdots, N$
N	The number of DC converters
a_{ij}	Adjacency element between the i-th and j-th DC converter
L	Laplacian matrix of a graph
V_B	DC bus voltage
R_i, L_i, C_i	Resistance, inductance and capacity
Symbols in primary control layer	
V_i,I_i	DC converter output voltage and current, respectively
V_i^{ref}	Reference voltage for V_i
I_i^{ref}	Reference current for I_i
V^*	Nominal DC bus voltage
k_i	Droop gain
K_{VPi},K_{VIi}	PI control parameters in voltage control loop
K_{IPi},K_{IIi}	PI control parameters in current control loop
Symbols in secondary control layer	
u_i	Secondary control signal
e	DC bus voltage error
$\bar{V}_i$	Virtual voltage drop
$\bar{V}_{avg}$	$\bar{V}_{avg} \triangleq \frac{1}{N}\sum_{i=1}^{N}\bar{V}_i$
x_i	Intermediate variable in distributed secondary controller
α,β	Design parameters in distributed secondary controller
$\tilde{e}_i$	$\tilde{e}_i \triangleq u_i - \bar{V}_{avg}$
$\tilde{e}$	$\tilde{e} \triangleq [\tilde{e}_1, ..., \tilde{e}_N]^T$
$\bar{V}$	$\bar{V} \triangleq [\bar{V}_1, ..., \bar{V}_N]^T$
$\tilde{e}_T$,A,B,S,γ	Intermediate variables for stability analyses
τ	Time delay

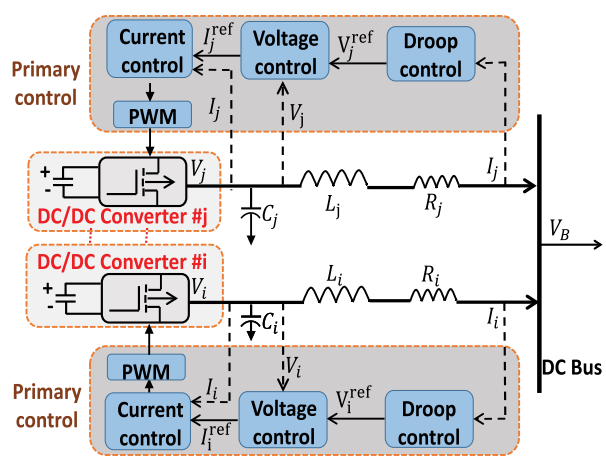

Fig. 1. Decentralized droop control.

II. Problem Formulation and Control Objective

As shown in Fig. 1, this paper considers a DC microgrid with N parallel DC/DC converters connected to a DC bus [23], [24]. Traditionally, the load current sharing among the DC sources is achieved by decentralized droop control. In this section, we will briefly review the droop control method and point out its limitations in voltage restoration. After that, we will present our control objective.

A. Droop Control

For converter #i, $i = 1, \ldots, N$, by properly designing its voltage and current control loops, its output voltage V_i can well track its reference voltage V_i^{ref} [24], i.e., $V_i = V_i^{ref}$. To achieve proper current sharing among the converters, the voltage reference V_i^{ref} is generated by a droop control function

$$V_i^{ref} = V^* - k_i I_i \tag{1}$$

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where V^* is the nominal DC voltage in open-loop, k_i denotes the droop gain, and I_i is the output current of converter #i.

The DC bus voltage V_B is computed as $V_B = V_i - R_i I_i$ where R_i is the resistance of power transmission line i. Then we obtain

$$V_B = V^* - (R_i + k_i)I_i \tag{2}$$

From (2), it can be obtained that

$$(R_i + k_i)I_i = \big(R_j + k_j\big)I_j, \quad \forall i, j = 1, \ldots, N \tag{3}$$

Therefore, the current sharing ratio among the converters is related to the sum of the droop gain k_i and line resistance R_i, i.e., $I_i/I_j = (R_j + k_j)/(R_i + k_i)$. Typically, the droop gain k_i is set much larger than the line resistance R_i, i.e., $k_i \gg R_i$, thus we have $I_i/I_j = k_j/k_i$.

As discussed above, by setting a proper droop gain, proportional current sharing among the converters can be achieved. The droop gain can be seen as a "virtual resistance" [26].

B. Control Objective

For proper current sharing, the droop gain k_i should be set much larger than the line resistance R_i. However, as shown in (2), a larger k_i gives a larger deviation between the bus voltage V_B and its nominal voltage V^*.

Therefore, the control objective of this paper is to tackle the droop control limitations by introducing a distributed secondary control layer. Specifically, the new control scheme should ensure both proper current sharing and voltage restoration, i.e.,

$$\lim_{t\to\infty} I_i/I_j = k_j/k_i \tag{4}$$

$$\lim_{t\to\infty} e = V_B - V^* = 0. \tag{5}$$

Remark 1: The considered DC system in this paper has a typical DC microgrid structure. It has been widely investigated in the existing literature [18]–[25]. Due to its simplicity, this structure has also been applied in practical systems such as the More Electric Aircraft [27].

C. Data Communication Network

As part of the fundamental infrastructure in power systems, communication networks play an important role in system monitoring and control. The distributed control to be developed in this paper for current sharing and voltage restoration in DC microgrid relies on communications among DC converters.

A communication network is denoted as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where $\mathcal{V}$ denotes the set of the converters, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the communication edge set, and $\mathcal{A} \in \mathbb{R}^{N\times N}$ is the adjacency matrix. Let a_{ij} denote the entities of $\mathcal{A}$. Then, $a_{ij} = 1$ if $(\upsilon_j, \upsilon_i) \in \mathcal{E}$, and $a_{ij} = 0$ if $(\upsilon_j, \upsilon_i) \notin \mathcal{E}$.

The Laplacian matrix L of graph $\mathcal{G}$ is formed with elements $L_{ii} = \sum_{i\neq j} a_{ij}$, and $L_{ij} = -a_{ij}$ for $i \neq j$. Define $Sym(L) = \frac{1}{2}(L + L^T)$, and let $\hat{\lambda}_1, \ldots, \hat{\lambda}_N$ denote the eigenvalues of $Sym(L)$. Then, if $\mathcal{G}$ is strongly connected and weight-balanced, zero is a simple eigenvalue of both L and $Sym(L)$. Without loss of generality, we denote $0 = \hat{\lambda}_1 < \hat{\lambda}_2 \leq \hat{\lambda}_3 \leq \cdots \leq \hat{\lambda}_N$, where $\hat{\lambda}_i$ is the ith eigenvalue.

In this paper, the communication graph among the converters is assumed to be strongly connected and weight-balanced. Since the communication channel between two converters is two-directional, this requirement is easy to meet as long as all converters are connected.

III. Distributed Secondary Control

This section presents a new distributed secondary control scheme to achieve the control objective (4)-(5). As shown in Fig. 1, each converter is directly controlled by a droop control function. Then, a secondary control signal u_i is to be designed to add into the droop function (1) in the primary control layer, i.e.,

$$V_i^{ref} = V^* - k_i I_i + u_i \tag{6}$$

Then, we have

$$V_B = V^* - (k_i + R_i)I_i + u_i \tag{7}$$

Now we are ready to deign the secondary controller u_i.

A. Secondary Controller With Global Information

Before giving the distributed secondary controller, we will first present a secondary controller when global information of the converters is available. This facilitates our development later for distributed secondary controller without global information.

Firstly, a new parameter 'virtual voltage drop' is defined as

$$\bar{V}_i = (k_i + R_i)I_i, \quad i = 1, \ldots, N \tag{8}$$

which is time-varying. When $\bar{V}_i$ is globally available, the secondary controller can be designed as

$$u_i = \frac{1}{N}\sum_{i=1}^{N} \bar{V}_i = \bar{V}_{avg}, \quad i = 1, \ldots, N \tag{9}$$

It is seen from (9) that the secondary control signal u_i is the average value of $\bar{V}_i$, $i = 1, \ldots, N$. With this controller, we have the following results in Theorem 1.

Theorem 1: With the controller (8) and (9), the control objective (4)-(5) can be achieved.

Proof: By substituting (8) and (9) into (7) gives

$$e = V_B - V^* = \bar{V}_{avg} - \bar{V}_i, \quad i = 1, \ldots, N \tag{10}$$

From (10), we have $Ne = N\bar{V}_{avg} - \sum_{i=1}^{N} \bar{V}_i = 0$. Therefore, (5) is achieved. Meanwhile, since $e = 0$, from (10) it is obtained that

$$\bar{V}_i = \bar{V}_j = \bar{V}_{avg}, \quad \forall i, j = 1, \ldots, N \tag{11}$$

Since $k_i \gg R_i$, from (8) and (11) we have $I_i/I_j = k_j/k_i$. Thus, (4) is ensured. This completes the proof. ■

Remark 2: As in [18], [23], [24], it is assumed that the line resistances are known. When there exists a line resistance measurement error $\Delta R_i = R_i - R_{i,real}$, it is easy to obtain that in the steady state the voltage violation error is $V_B - V^* = \frac{1}{N}\sum_{i=1}^{N} I_i \Delta R_i$. Since the measurement error ΔR_i

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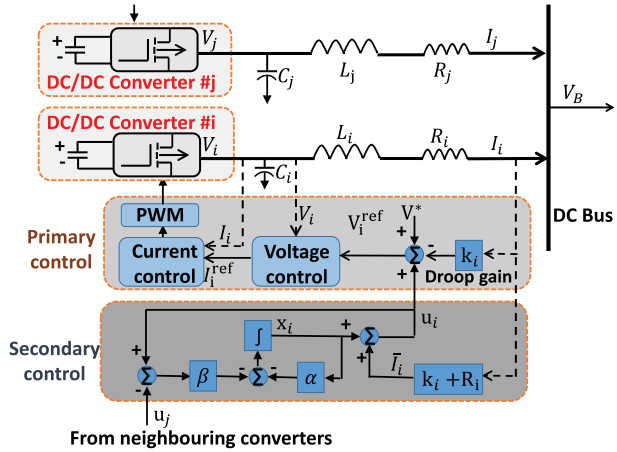

Fig. 2. Distributed secondary control diagram.

is small, the effects of these line resistance measurement errors on the control performance are negligible.

Remark 3: It is seen from (8) that the newly defined parameter $\bar{V}_i$ has a clear physical meaning. It represents the 'virtual voltage drop' caused by the droop gain k_i and line resistance R_i. As shown in Theorem 1, by compensating for this part in the secondary control layer, both voltage restoration and current sharing can be achieved.

Remark 4: The secondary controller (8)-(9) requires the global information of $\bar{V}_i$ in the computation of $\bar{V}_{avg}$. While such global information may be accessible in small-scale systems, it is generally unavailable in large-scale systems with a large number of converters, causing difficulties in implementation of the controller. This motivates our work in this paper on distributed secondary control scheme.

B. Distributed Secondary Controller

The key idea of the secondary controller with global information is to use the average value $\bar{V}_{avg}$ as the control signal, as seen in (9). However, when the global information $\bar{V}_i$, $i = 1, \ldots, N$, is not available, (9) cannot be implemented. To solve this problem, we design a distributed method to enable each converter to estimate $\bar{V}_{avg}$ by only exchanging information with its neighbours. Inspired by the dynamic average consensus algorithms in [29], [30], a distributed secondary controller is designed as

$$\begin{cases} \dot{x}_i = -\alpha x_i - \beta \Sigma_{j\in N_i} a_{ij}(u_i - u_j) \\ u_i = x_i + \bar{V}_i, \quad i = 1, \ldots, N \end{cases} \tag{12}$$

The control structure for the i-th converter is shown in Fig. 2. As seen in Fig. 2, the information required to be exchanged among the DC converters is only the control input signal u_i. Moreover, the proposed secondary controller does not require the DC bus voltage as the feedback signal. Therefore, it is simple and easy to implement. We have the following results in Theorem 2.

Theorem 2: With the controller (12), when the communication graph among the converters is strongly connected and weight-balanced, the control objective (4)-(5) can be achieved.

Proof: From (7), we have

$$e = u_i - \bar{V}_i, \quad i = 1, \ldots, N \tag{13}$$

Adding (13) together yields

$$Ne = \sum_{i=1}^{N} u_i - \sum_{i=1}^{N} \bar{V}_i = \sum_{i=1}^{N} (u_i - \bar{V}_{avg}) = \sum_{i=1}^{N} \tilde{e}_i \tag{14}$$

where $\tilde{e}_i = u_i - \bar{V}_{avg}$. Then, from (12) and (13), we have

$$\begin{aligned} \dot{\tilde{e}}_i &= \dot{\bar{V}}_i - \dot{\bar{V}}_{avg} - \alpha(u_i - \bar{V}_i) - \beta \Sigma_{j\in N_i} a_{ij}(\tilde{e}_i - \tilde{e}_j) \\ &= -\alpha(u_i - \bar{V}_{avg}) + \dot{\bar{V}}_i - \dot{\bar{V}}_{avg} + \alpha \bar{V}_i - \alpha \bar{V}_{avg} \\ &\quad - \beta \Sigma_{j\in N_i} a_{ij}(\tilde{e}_i - \tilde{e}_j) \end{aligned} \tag{15}$$

Define $\tilde{e} = [\tilde{e}_1, \ldots, \tilde{e}_N]^T$, $\bar{V} = [\bar{V}_1, \ldots, \bar{V}_N]^T$, and $\Pi_N = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$. We have

$$\dot{\tilde{e}} = -\alpha\tilde{e} - \beta L\tilde{e} + \Pi_N\left(\dot{\bar{V}}_i + \alpha\bar{V}_i\right) \tag{16}$$

Now, consider the following coordination transformation

$$\tilde{e}_T = \begin{bmatrix} \tilde{e}_{T1} \\ \tilde{e}_{T2} \end{bmatrix} = \underbrace{\begin{bmatrix} \chi^T \\ S^T \end{bmatrix}}_{B} \tilde{e} \tag{17}$$

where $\chi = \frac{1}{\sqrt{N}}\mathbf{1}_{N\times 1}$, S is a matrix making $[\frac{1}{\sqrt{N}}\mathbf{1}_{N\times 1} \quad S]$ an orthonormal matrix. We have $S^TS = \mathbf{I}_{N-1}$, $SS^T = \Pi_N$ and $BB^T = \mathbf{I}_N$.

Since $\chi^T L = 0$ and $\chi^T \Pi_N = 0$, from (17) and (16), we have $\dot{\tilde{e}}_{T1} = -\alpha\tilde{e}_{T1}$ and

$$\dot{\tilde{e}}_{T2} = -(\beta S^T LS + \alpha I_{N-1})\tilde{e}_{T2} + S^T\Pi_N\left(\dot{\bar{V}}_i + \alpha\bar{V}_i\right) \tag{18}$$

Thus, we obtain $|\tilde{e}_{T1}| \le |\tilde{e}_{T1}(0)|\mathbf{e}^{-\alpha t}$ and

$$\tilde{e}_{T2} = \Phi(t,0)\tilde{e}_{T2}(0) + \int_0^t \Phi(t,s)S^T\Pi_N\left(\dot{\bar{V}}_i + \alpha\bar{V}_i\right)ds \tag{19}$$

where $\Phi(t,s) = \mathbf{e}^{-(\beta S^T LS + \alpha I_{N-1})(t-s)}$. Define $A = -(\beta S^T LS + \alpha I_{N-1})$. λ_A is the eigenvalue of $\frac{1}{2}(A + A^T)$. Then,

$$\begin{aligned} &\left|\lambda_A I_{N-1} - \frac{1}{2}(A + A^T)\right| \\ &\quad = \left|(\lambda_A + \alpha)I_{N-1} + \beta S^T LS\right| = 0 \end{aligned} \tag{20}$$

Since the eigenvalues of $S^T LS$ are $\hat{\lambda}_i$, $i = 2, \ldots, N$, it has $\lambda_A = -\alpha - \beta\hat{\lambda}_i$. From [31], $\Phi(t,s) \le \mathbf{e}^{(-\alpha-\beta\hat{\lambda}_2)(t-s)}$. Since both $\dot{\bar{V}}_i$ and $\bar{V}_i$ are bounded, we have $\|S^T\Pi_N(\dot{\bar{V}}_i + \alpha\bar{V}_i)\| \le \|\Pi_N(\dot{\bar{V}}_i + \alpha\bar{V}_i)\| = \gamma$, where γ is a bounded constant.

Therefore, we have

$$\|\tilde{e}_{T2}(t)\| \le \mathbf{e}^{\left(-\alpha-\beta\hat{\lambda}_2\right)t}\|\tilde{e}_{T2}(0)\| + \frac{\gamma}{\beta\hat{\lambda}_2}\left(1 - \mathbf{e}^{\left(-\alpha-\beta\hat{\lambda}_2\right)t}\right)$$

Thus, $\lim_{t\to\infty}\|\tilde{e}_{T2}(t)\| \le \frac{\gamma}{\beta\hat{\lambda}_2}$. Since $\|\tilde{e}\| = \tilde{e}_T^T BB^T\tilde{e}_T^T = \|\tilde{e}_T\|$, we obtain

$$\lim_{t\to\infty}\|\tilde{e}\| \le \lim_{t\to\infty}|\tilde{e}_{T1}| + \lim_{t\to\infty}\|\tilde{e}_{T2}\| \le \frac{\gamma}{\beta\hat{\lambda}_2} \tag{21}$$

As can be seen from (21), by increasing β, the bound of $\|\tilde{e}\|$ can be made arbitrarily small, i.e.,

$$\lim_{t\to\infty}|\tilde{e}_i| \le \lim_{t\to\infty}\|\tilde{e}\| \approx 0, \quad i = 1, \ldots, N \tag{22}$$

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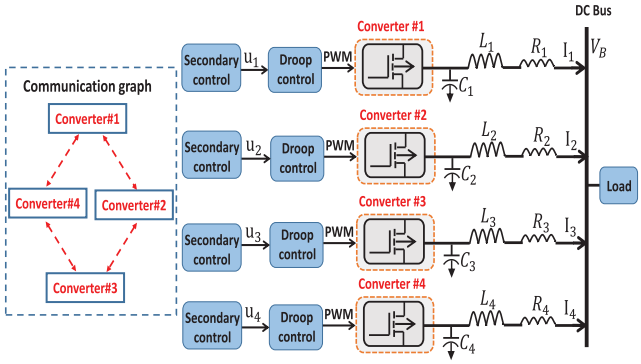

Fig. 3. Case study setup.

From (13), (14) and (22), it is obtained that $\lim_{t\to\infty} e = 0$. Thus, (5) is achieved. Meanwhile, it can also be obtained that

$$\lim_{t\to\infty} \bar{V}_i = \bar{V}_j = \bar{V}_{avg}, \quad \forall i, j = 1, \ldots, N \tag{23}$$

Therefore, (4) is achieved. This completes the proof. ■

Remark 5: A pioneer work in [24] proposes a distributed secondary control scheme for both power allocation and voltage restoration in DC microgrid. However, the control scheme in [24] requires to measure both current I_i and bus voltage V_B as feedback signals. In comparison, the proposed scheme in this paper only needs to measure I_i. Thus, less measurement and communication support is required. Moreover, it is not easy to set the control parameters in [24] as these parameters need to satisfy some specific stability conditions (see Lemma 2 and Theorem 1 in [24]), which are difficult to check. Therefore, the control scheme in this paper is simpler and easier to implement.

Remark 6: As can be seen from Theorem 2 and its proof, by increasing α and β, the control performance will become better. However, a bigger α and β will make the algorithm more sensitive to measurement errors, disturbances or time delay. Therefore, there exists a trade-off between the control performance and algorithm robustness in setting α and β. Moreover, in contrast to [24], the proposed controller (12) in this paper poses no requirement for the loads. Therefore, it is able to deal with different types of loads including resistance loads and CPLs. This will be illustrated later through case studies and experiments in Sections IV and V.

IV. Case Studies

The effectiveness of the proposed distributed secondary control scheme is demonstrated through five case studies in this section. The first case study examines the control scheme with resistance load, while the second case tests the CPLs. The plug-in and plug-out capability of the proposed scheme is tested in Case study 3. A comparison between the proposed scheme and the schemes in [22] and [24] is conducted in Case study 4. After that, the effects of our scheme on controlling multiple buses are tested in Case study 5.

Our case study model is built in MATLAB Simulink environment. As shown in Fig. 3, in these case studies, four DC/DC bulk converters are connected to a DC bus. In the droop control layer, two standard PI controllers are used in the voltage control loop and current control loop, respectively. To protect the electronic interfaces of the converters from abrupt damages, a filter is added to the secondary control output and the local voltage set point. Following [24], the system parameters and the droop control parameters are given in Table II. The nominal DC bus voltage is set as $V^* = 48V$.

TABLE II
System and Primary Control Parameters [24]

	DC1	DC2	DC3	DC4
DC Parameters	V_{DC1}=100 V f_{s1}=1.25 kHz	V_{DC2}=100 V f_{s2}=1.25 kHz	V_{DC3}=100 V f_{s3}=1.25 kHz	V_{DC4}=100 V f_{s4}=1.25 kHz
Voltage Loop	K_{VP1}=4 K_{VI1}=800	K_{VP2}=4 K_{VI2}=800	K_{VP3}=4 K_{VI3}=800	K_{VP4}=4 K_{VI4}=800
Current Loop	K_{IP1}=5 K_{II1}=110	K_{IP2}=5 K_{II2}=110	K_{IP3}=5 K_{II3}=110	K_{IP4}=5 K_{II4}=110
LC Filter	L_1=0.01H C_1=2200μf	L_2=0.01H C_2=2200μf	L_3=0.01H C_3=2200μf	L_4=0.01H C_4=2200μf
Line Resistance	R_1=0.01Ω	R_2=0.01Ω	R_3=0.01Ω	R_4=0.01Ω
Droop Gain	K_1=6	K_2=3	K_3=2	K_4=2

A. Case Study 1: Resistance Load

This case study examines the control effects of the proposed scheme with resistance load. The secondary control parameters in (12) are set as $\alpha = 50$ and $\beta = 50$. It is assumed that the DC system goes through the following 4 stages:

Stage 1 (0-2s): A resistance load $R_{L1} = 5\Omega$ is connected at $t = 0s$ but only primary control is activated;

Stage 2 (2-4s): Secondary control is implemented at $t = 2s$;

Stage 3 (4-6s): Another resistance load $R_{L2} = 5\Omega$ is connected to the grid at $t = 4s$;

Stage 4 (6-8s): R_{L2} is disconnected from the grid at $t = 6s$;

Case study results are shown in Fig. 4. From Fig. 4(b), it is observed that in the first two seconds only droop control is employed, the DC bus voltage will drop to around $42V$, deviating severely from the desired value $48V$. After the distributed secondary controller is activated at $t = 2s$, the bus voltage is quickly restored to $48V$. Moreover, when the load connected to the bus changes, the bus voltage can still be well maintained. From Fig. 4(a), it is observed that the presented scheme can also ensure proper current sharing as desired, regardless of the load changes. These case study results illustrate that the proposed scheme guarantees both proper current sharing and voltage restoration.

B. Case Study 2: Constant Power Loads

This case tests our proposed scheme with CPLs. The parameters are set the same as in Case 1, while the system goes through the following five stages:

Stage 1 (0−2s): A CPL $P_{L1} = 500W$ is connected and only primary control is activated at $t = 0s$;

Stage 2 (2−4s): Secondary control is implemented at $t = 2s$;

Stage 3 (4−6s): Another CPL $P_{L2} = 200W$ is connected to the grid at $t = 4s$;

Stage 4 (6−8s): P_{L2} is disconnected from the grid at $t = 6s$;

Our case study results are shown in Fig. 5. It is seen from Fig. 5 that with a CPL the primary droop control cannot maintain the bus voltage at the nominal value. While

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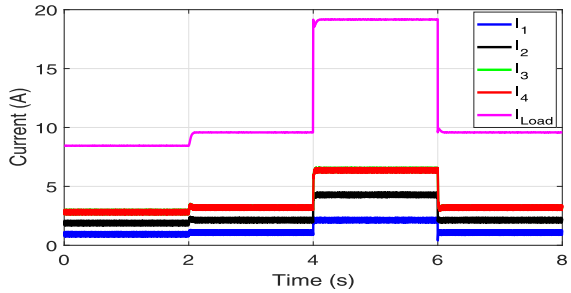
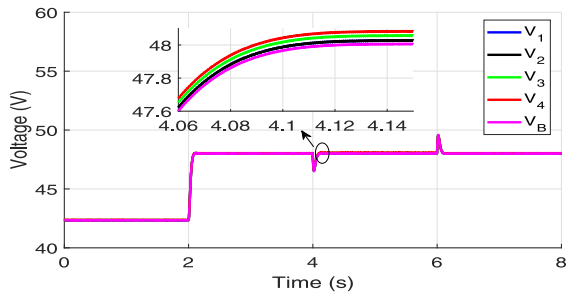
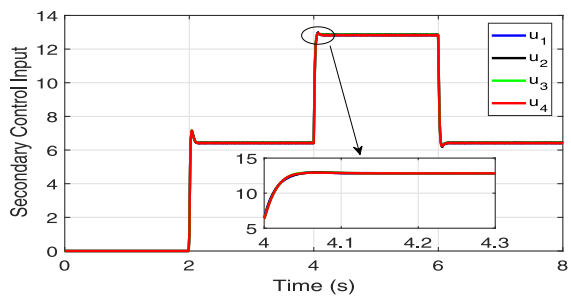


(a) Current output

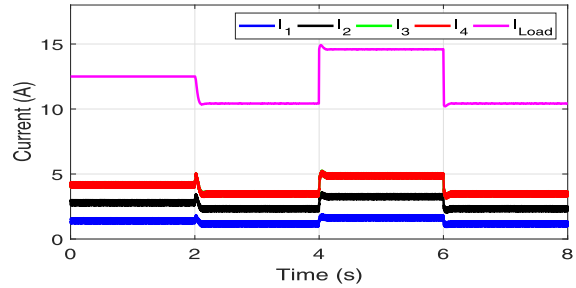


(b) Voltage output

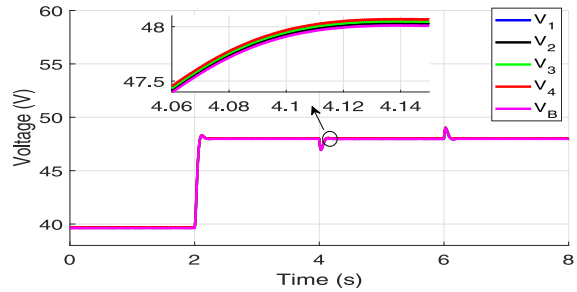


(c) Secondary control signal.

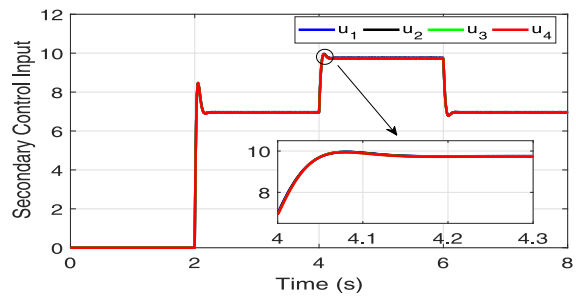
Fig. 4. Resistance load.



(a) Current output



(b) Voltage output



(c) Secondary control signal

Fig. 5. Constant power load.

with our proposed secondary control scheme, though the CPL keeps changing, the DC bus voltage is well maintained with proper current sharing among all inverters. This case study shows the capability of the proposed scheme in dealing with CPLs.

C. Case Study 3: Plug-In and Plug-Out Capability

In this case, the plug-in and plug-out capability of the proposed control scheme is tested. The design parameters are set the same as in Case 1. This case study goes through the following 3 stages:

Stage 1 (0-2s): At $t = 0s$, $R_{L1} = 5\Omega$ is connected, and both primary and secondary control are activated;

Stage 2 (2-4s): Converter #4 is disconnected from the grid at $t = 2s$;

Stage 3 (4-6s): Converter #4 is connected to the grid again at $t = 4s$;

The case study results are shown in Fig. 6. Fig. 6(a) shows the current performance, while the DC bus voltage and the secondary control signal for each converter is shown in Fig. 6(b) and Fig. 6(c), respectively. It is observed that when converter #4 is disconnected from the grid, the current can be shared quickly as desired among the remaining three converters. When converter #4 is re-connected to the grid again, the current is properly shared again among all the four converters. Moreover, regardless of the connection and disconnection, the DC bus voltage can be well maintained at the nominal value. This demonstrates the plug-in and plug-out capability of the proposed control scheme.

D. Case Study 4: Comparison With the Existing Schemes

This case study compares the proposed control scheme in this paper with the ones presented in [22] and [24]. It is assumed that this case study goes through the same stages as in Case 1. The parameters for our scheme is set as $\alpha = 5$, $\beta = 10$. The parameters for [22] is set as $K_p = 2$ and $K_i = 0.1$. The parameters for [24] are set as $\alpha = 1$, $\beta = 1$, $K_p = 1$ and $K_I = 40$. In this case study, we assume that no converter is able to access to the DC bus voltage.

The case study results are shown in Fig. 7 and Fig. 8. It is seen that when no converter has access to the bus voltage, the control scheme in [24] fails to maintain the bus voltage, and there exists a stable voltage violation error for [22]. Moreover, before the secondary is implemented, the scheme in [22] fails

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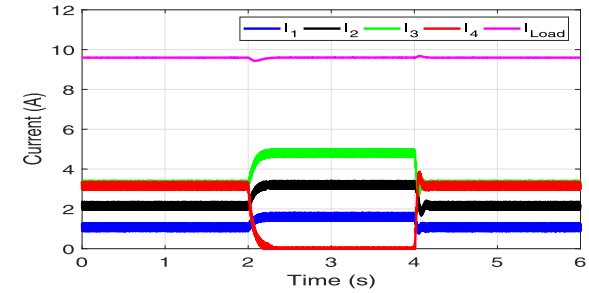
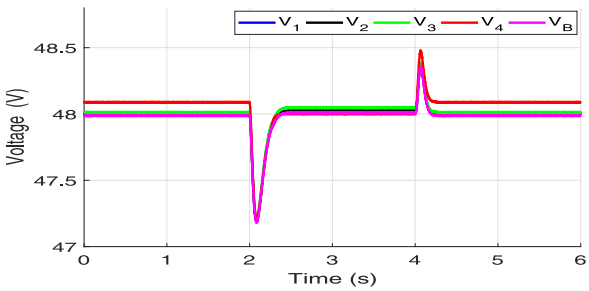
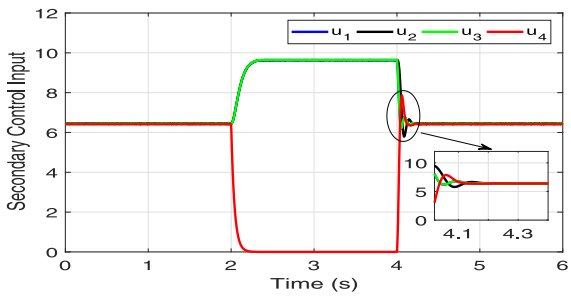


(a) Current output

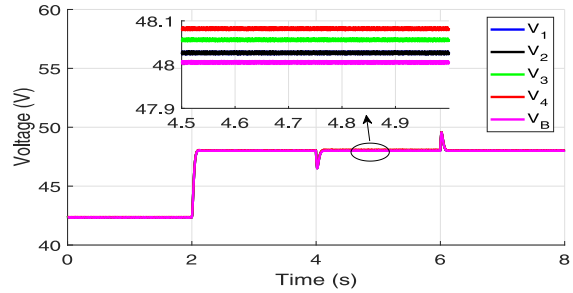


(b) Voltage output

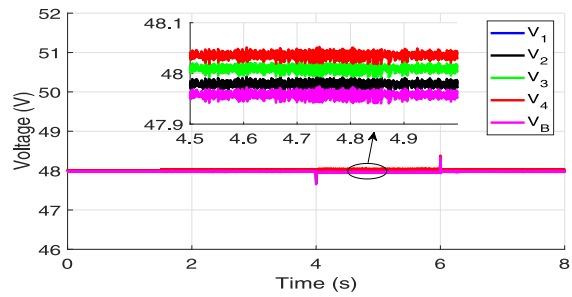


(c) Secondary control signal

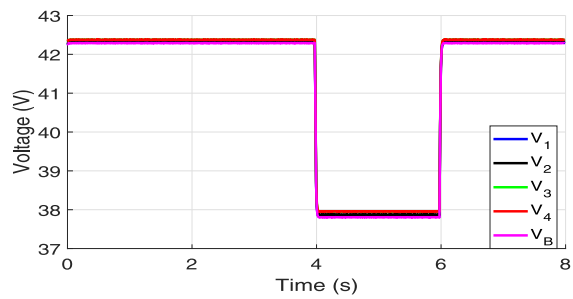
Fig. 6. Plug-in and plug-out capability.



(a) Voltage output with the proposed scheme.



(b) Voltage output with the scheme in [22].



(c) Voltage output with the scheme in [24].

Fig. 7. DC bus voltage.

to ensure the desired current sharing. In contrast, our control scheme maintains the bus voltage and current sharing very well. This clearly illustrates the advantages of our scheme compared with the ones in [22] and [24].

E. Case Study 5: Control Performance With Multiple Buses

In this case, we test our scheme for a DC microgrid with two buses, as shown in Fig. 9. We consider an extreme case where the load is only connected to Bus #1. It is assumed that this system goes through the same stages as in Case study 1. The resistance of the transmission line connecting these two buses is set as $R_L = 0.05\Omega$.

Our case study results are shown in Fig. 10. When the load is connected, the current will flow from Bus #2 to Bus #1, causing a voltage violation between V_{B1} and V_{B2}. However, it is observed from Fig. 10 that, with our control scheme, both V_{B1} and V_{B2} stay close to the reference value 48V, and their average value equals to 48V. Moreover, the desired current sharing is also guaranteed. This demonstrates the capability of our propsoed scheme in controlling DC microgrid with more complicated topologies.

TABLE III
PARAMETERS OF THE EXPERIMENTAL SETUP

	DC1	DC2	DC3
DC Parameters	V_{DC1}=50 V f_{s1}=20 kHz	V_{DC2}=50 V f_{s2}=20 kHz	V_{DC3}=50 V f_{s3}=20 kHz
Voltage Loop	K_{VP1}=0.5 K_{VI1}=10	K_{VP2}=0.5 K_{VI2}=10	K_{VP3}=0.5 K_{VI3}=10
Current Loop	K_{IP1}=1 K_{II1}=60	K_{IP2}=1 K_{II2}=60	K_{IP3}=1 K_{II3}=60
LC Filter	L_1=2mH C_1=470μF	L_2=2mH C_2=470μF	L_3=2mH C_3=470μF
Droop Gain	K_1=2	K_2=3	K_3=6
Line resistance	$R_1 = 0.1\Omega$	$R_2 = 0.1\Omega$	$R_3 = 0.1\Omega$
DC voltage reference	$V^* = 100V$		

V. EXPERIMENTAL RESULTS

In this section, the proposed scheme is verified by experiments over a DC MG with three DC/DC boost converters built in the laboratory. The experiment setup is shown in Fig. 11. The control scheme is implemented into a dSPACE1006 control platform to generate PWM signals for the three converters. The sampling and PWM frequency are set the same as 20KHz. The secondary control parameters are set as $\alpha = 5$ and $\beta = 10$. The other control parameters for the experiment setup are summarized in Table III.

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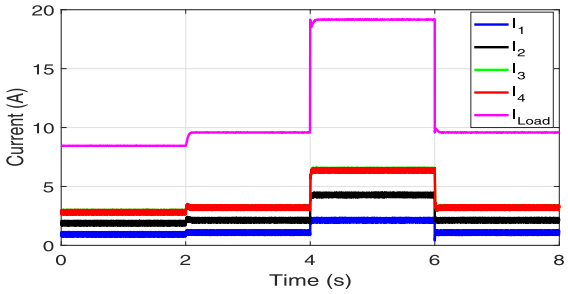
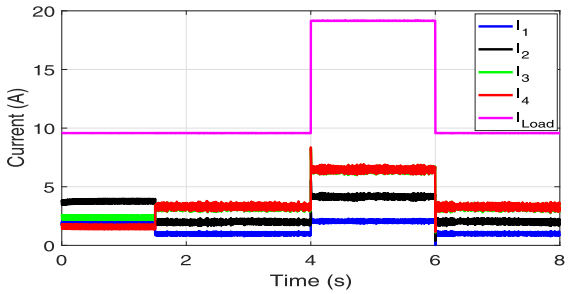
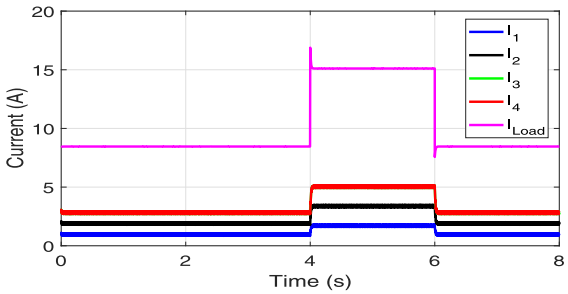


(a) Current output with the proposed scheme.



(b) Current output with the scheme in [22].



(c) Current output with the scheme in [24].

Fig. 8. Current output.

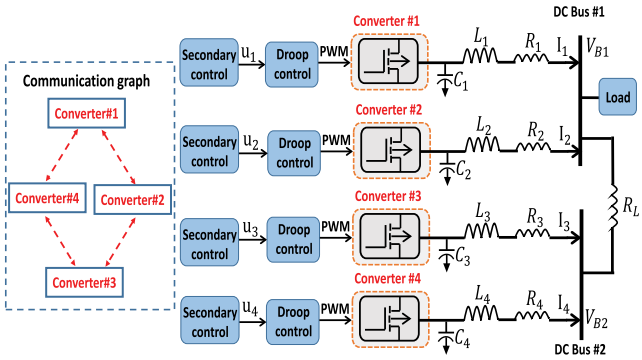
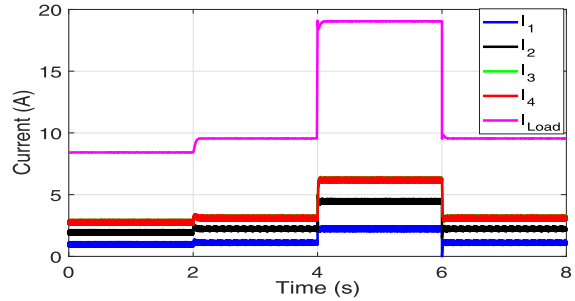
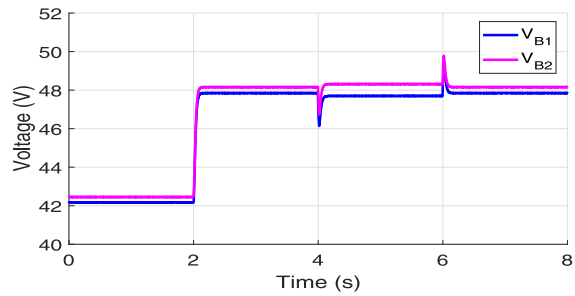


Fig. 9. Case study setup with two buses.



(a) Current output



(b) Voltage output

Fig. 10. Control performance with two buses.

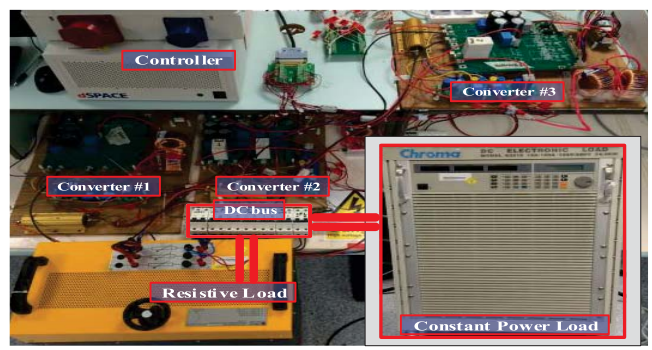


Fig. 11. Experiment setup.

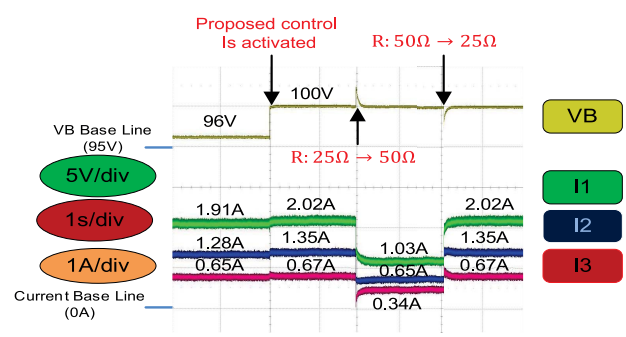


Fig. 12. Resistance load change.

A. Experimental Results With Resistance Load

In this case, initially only a resistance load $R = 25\Omega$ is connected to the bus and the secondary control scheme is not activated. As can be observed in Fig. 12, the DC bus voltage drops to $96V$, deviating from the nominal voltage reference. After our proposed secondary control scheme is activated, the DC bus voltage quickly goes back to the reference value $100V$. In order to test our control scheme under changing load files, the resistance load changes from 25Ω to 50Ω and then goes back to 25Ω. It is observed from Fig. 12 that, despite the change of loads, the bus voltage is well maintained at the nominal value. The current among the three converters are also properly shared as designed. These experimental results are consistent with our theoretical analyses.

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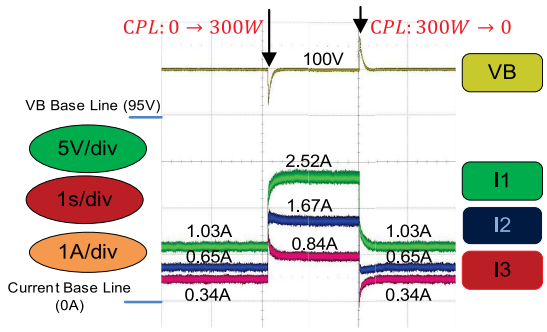


Fig. 13. Constant power load change.

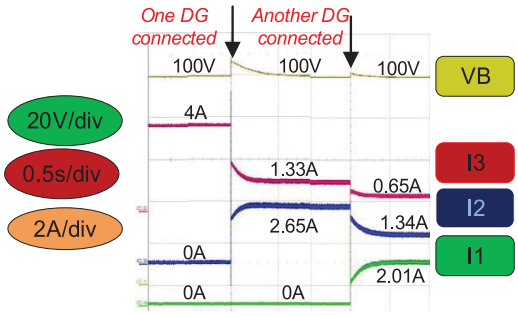


Fig. 14. Plug-and-play capability.

B. Experimental Results With CPLs

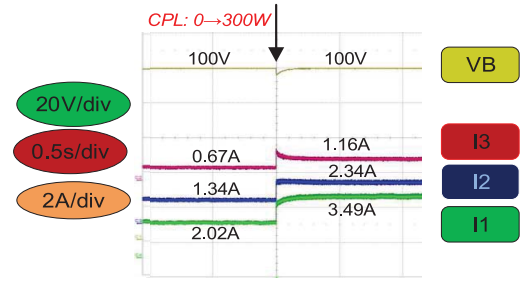
In this case, we test our control scheme with CPLs. For experiment, initially a resistance load $R = 50\Omega$ is connected and the secondary control scheme is also activated. A CPL load $P = 300W$ is connected and then disconnected to the bus. Our experiment results are shown in Fig. 13. It is observed that, when the CPL is added, the bus voltage can quickly go back to the reference value, and the proper current sharing is also guaranteed. After the disconnection of the CPL, both the voltage and current return to their previous values before the CPL is added. Again, these experiment results illustrate the effectiveness of our proposed scheme.

C. Experimental Results With Plug-and-Play

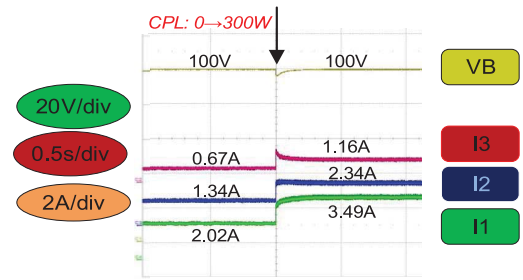
This case tests the plug-and-play capability of our control scheme. In experiment setup, a resistance load $R = 25\Omega$ is connected and the secondary control scheme is also activated. Initially, only one DC converter is connected to the bus, and then the second one is connected, and at last the third is connected. Our experiment results are shown in Fig. 14. It is shown that, when new DC converters are connected to the DC bus, both the bus voltage and proper current sharing can be guaranteed.

D. Experimental Results With Time Delay

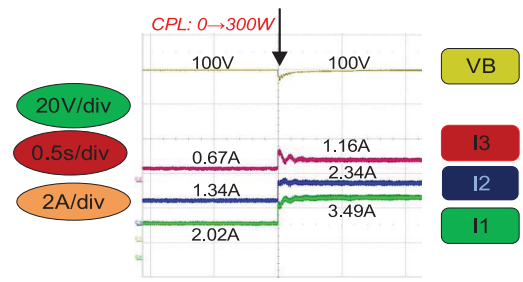
In this case, the time delay effects on our control scheme is tested. In this setup, initially a resistance load $R = 25\Omega$ is connected and the secondary control scheme is also activated. A CPL load $P = 300W$ is then connected to the bus. The experiment results with three different time delays, i.e., 1ms, 10ms and 50ms, are shown in Fig. 15. It is observed that, with a time delay up to 50ms, the control performance are still satisfactory. Therefore, it can be concluded that satisfactory control performance can be guaranteed with a time delay less than 50ms for this test example. Since for long-range low-delay network such as cellular communication network, the time delay is almost negligible, thus it is safe to implement our control scheme.



(a) Control performance with time delay as 1ms.



(b) Control performance with time delay as 10ms.



(c) Control performance with time delay as 50ms.

Fig. 15. Control performance with different time delays.

VI. Conclusion and Future Work

A new distributed secondary control scheme has been presented in this paper for a DC microgrid. It not only guarantees proper current sharing but also maintains the DC bus voltage at the nominal value. In implementation of the proposed scheme, the DC bus voltage is not required. Also, the proposed scheme is able to handle different types of loads. Case studies and experiments have demonstrated the effectiveness of the proposed scheme. Possible future work includes extending the results to systems with more complicated topologies, and theoretically analyzing the effects of time delay on the control performance.

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